

## Deducing oxidation number from formula

- Deduce the oxidation number of the underlined atom

|                                         |                                 |
|-----------------------------------------|---------------------------------|
| $\underline{\text{S}}\text{O}_4^{2-}$   | $x + 4(-2) = -2$<br>$x = +6$    |
| $\underline{\text{N}}\text{O}_3^-$      | $x + 3(-2) = -1$<br>$x = +5$    |
| $\text{Na}_2\underline{\text{O}}_2$     | $2(+1) + 2(x) = 0$<br>$x = -1$  |
| $\underline{\text{S}}_2\text{O}_3^{2-}$ | $2(x) + 3(-2) = -2$<br>$x = +2$ |

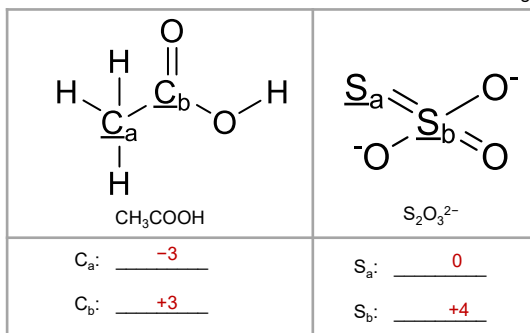
Note:

- oxidation number  $\equiv$  oxidation state

## Deducing oxidation number from structure

(to be covered in later topics)

- Deduce the oxidation number of the atoms in the following species



Notes:

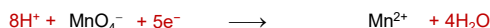
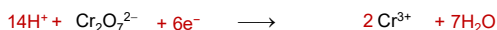
- this method allows us to deduce the specific oxidation number of each atom, instead of just the average oxidation number
- electronegativity increases from B < H < C, where C is the most electronegative

## Balancing half equations

### Acidic medium

- Balance the element being reduced or oxidised
- Balance O atoms by adding H<sub>2</sub>O
- Balance H atoms by adding H<sup>+</sup>
- Balance the overall charges by adding e<sup>-</sup>

- Balance the half equation for the following reactions in acidic medium



Note: common reduction half-equations can be found in *Data Booklet*

### Basic medium

- Balance the element being reduced or oxidised
- Balance O atoms by adding H<sub>2</sub>O
- Balance H atoms by adding H<sup>+</sup>
- Balance the overall charges by adding e<sup>-</sup>
- "Neutralise" H<sup>+</sup> ions by adding OH<sup>-</sup> ions to both sides of the half equation
- Combine the H<sup>+</sup> and OH<sup>-</sup> ions on same side of equation to form H<sub>2</sub>O
- Re-balance the half equation by eliminating any common terms, such as H<sub>2</sub>O

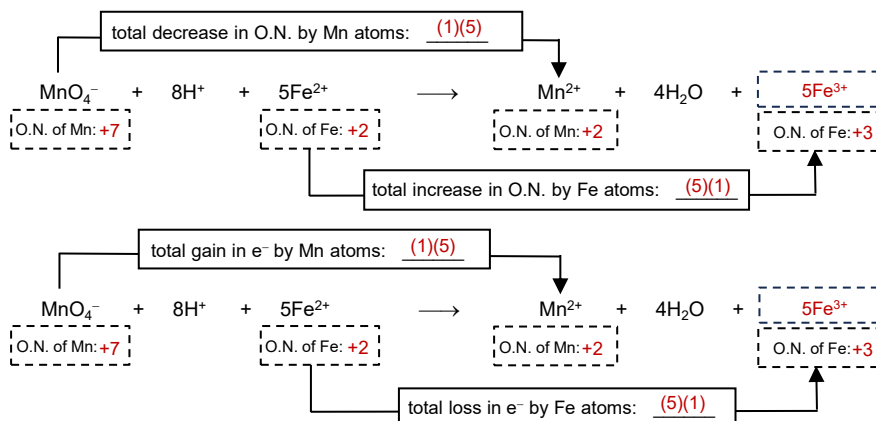
- Balanced the half equation for the following reaction in basic medium



## Summing up half equations

|                                |                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------|
| oxidation half-eq <sup>a</sup> | $2\text{I}^- \longrightarrow \text{I}_2 + 2\text{e}^-$                                           |
| reduction half-eq <sup>a</sup> | $2\text{IO}_3^- + 12\text{H}^+ + 10\text{e}^- \longrightarrow \text{I}_2 + 6\text{H}_2\text{O}$  |
| overall equation               | $2\text{IO}_3^- + 12\text{H}^+ + 10\text{I}^- \longrightarrow 6\text{I}_2 + 6\text{H}_2\text{O}$ |

## Solving by oxidation numbers OR e<sup>-</sup>



Note: when calculating total ↑ or ↓ in O.N or e<sup>-</sup>, must consider the number of atoms

## Deducing oxidation number from formula

- Deduce the oxidation number of the underlined atom

|                                         |                              |
|-----------------------------------------|------------------------------|
| $\underline{\text{S}}\text{O}_4^{2-}$   | $x + 4(-2) = -2$<br>$x = +6$ |
| $\underline{\text{N}}\text{O}_3^-$      |                              |
| $\text{Na}_2\underline{\text{O}}_2$     |                              |
| $\underline{\text{S}}_2\text{O}_3^{2-}$ |                              |

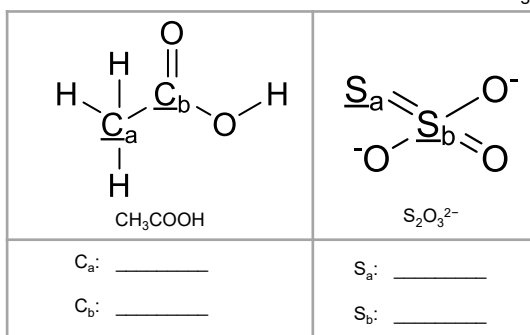
Note:

- oxidation number  $\equiv$  oxidation state

## Deducing oxidation number from structure

(to be covered in later topics)

- Deduce the oxidation number of the atoms in the following species



Notes:

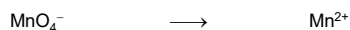
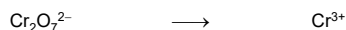
- this method allows us to deduce the specific oxidation number of each atom, instead of just the average oxidation number
- electronegativity increases from B < H < C, where C is the most electronegative

## Balancing half equations

### Acidic medium

- Balance the element being reduced or oxidised
- Balance O atoms by adding H<sub>2</sub>O
- Balance H atoms by adding H<sup>+</sup>
- Balance the overall charges by adding e<sup>-</sup>

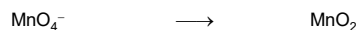
- Balance the half equation for the following reactions in acidic medium



### Basic medium

- Balance the element being reduced or oxidised
- Balance O atoms by adding H<sub>2</sub>O
- Balance H atoms by adding H<sup>+</sup>
- Balance the overall charges by adding e<sup>-</sup>
- "Neutralise" H<sup>+</sup> ions by adding OH<sup>-</sup> ions to **both** sides of the half equation
- Combine the H<sup>+</sup> and OH<sup>-</sup> ions on same side of equation to form H<sub>2</sub>O
- Re-balance the half equation by eliminating any common terms, such as H<sub>2</sub>O

- Balanced the half equation for the following reaction in basic medium

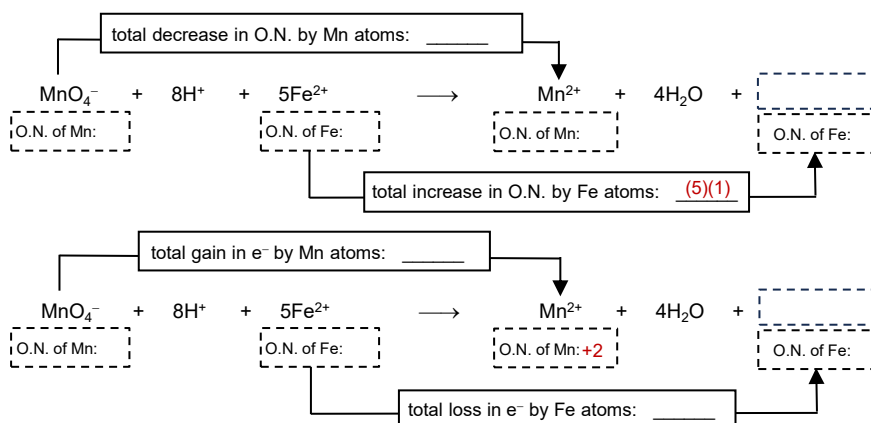


Note: common reduction half-equations can be found in *Data Booklet*

## Summing up half equations

|                                |                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------|
| oxidation half-eq <sup>a</sup> | $2\text{I}^- \longrightarrow \text{I}_2 + 2\text{e}^-$                                          |
| reduction half-eq <sup>a</sup> | $2\text{IO}_3^- + 12\text{H}^+ + 10\text{e}^- \longrightarrow \text{I}_2 + 6\text{H}_2\text{O}$ |
| overall equation               |                                                                                                 |

## Solving by oxidation numbers OR e<sup>-</sup>



Note: when calculating total ↑ or ↓ in O.N or e<sup>-</sup>, must consider the number of atoms